

FRANK PORTER
2016-2017

M1 MACRO

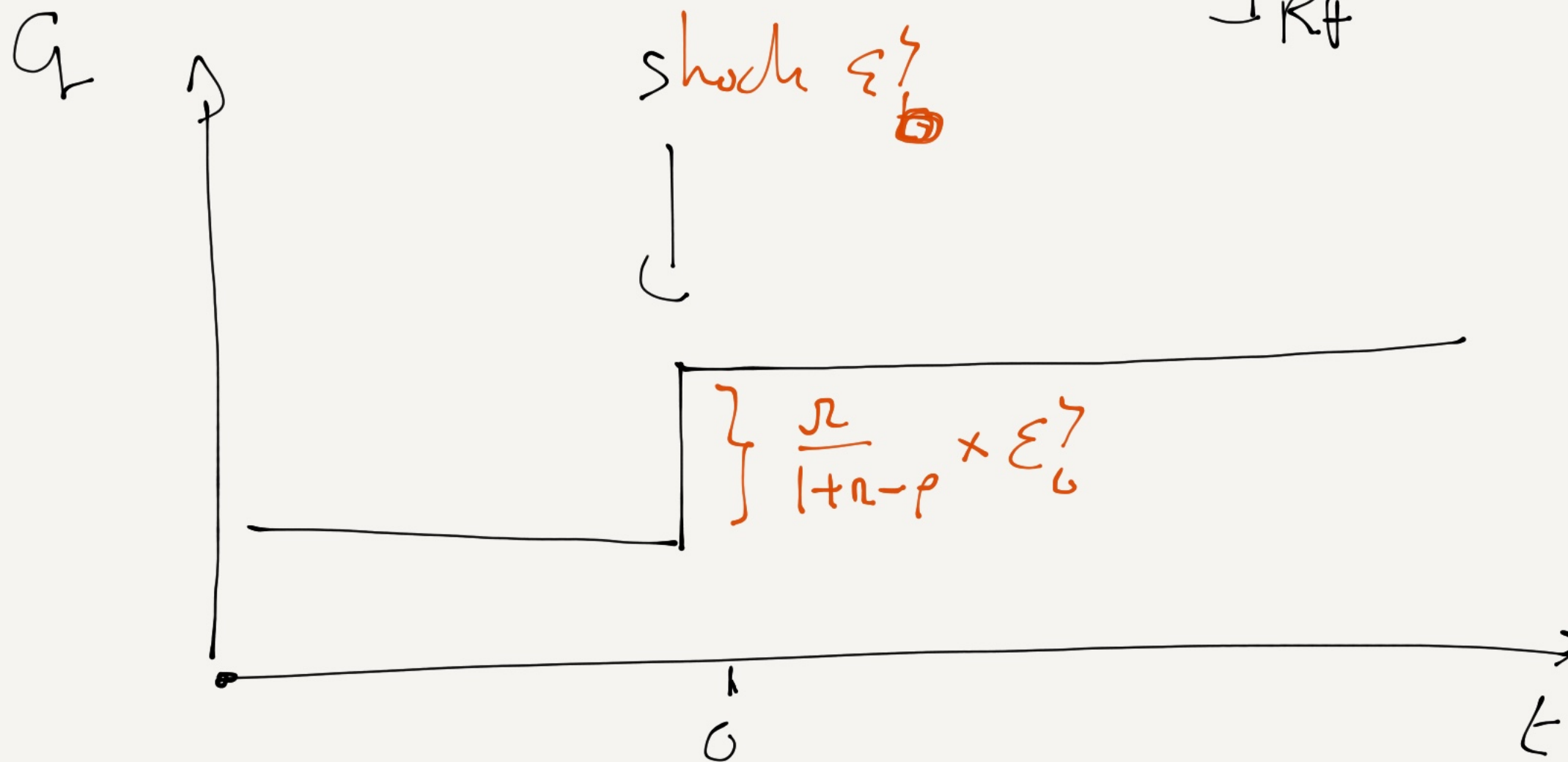
Group A

Session 12

These notes are written during class. They are posted "for the record" and may contain mistakes. They are not intended to be read by anyone that did not attend the class.

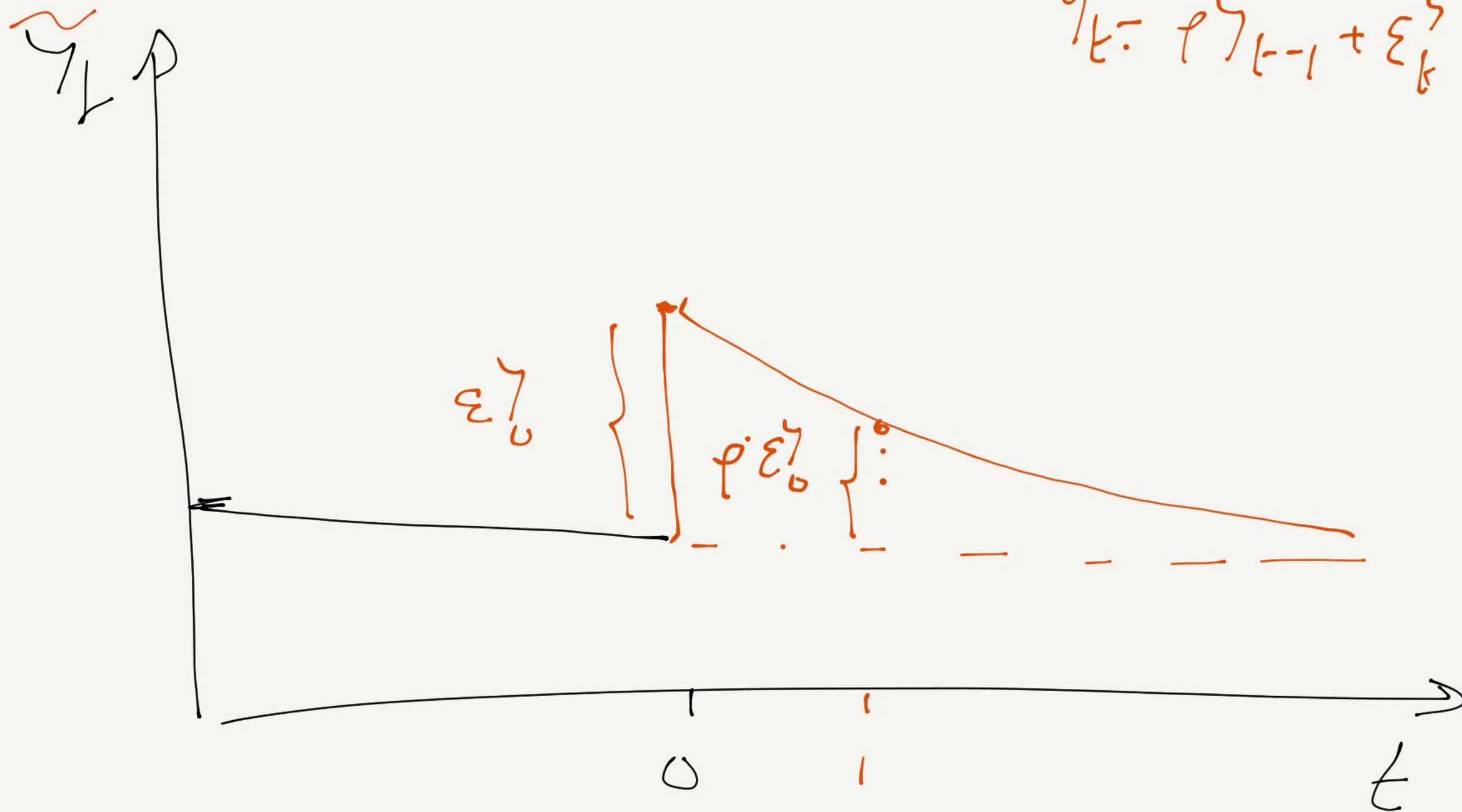
$$C_t = C_{t-1} + \frac{\alpha}{1+\alpha-\rho} \varepsilon_t^y$$

Response of C_t to ε_t^y (Impulse Response Function)
IRF



IRF of $\tilde{y}_t$ to ε_t^y

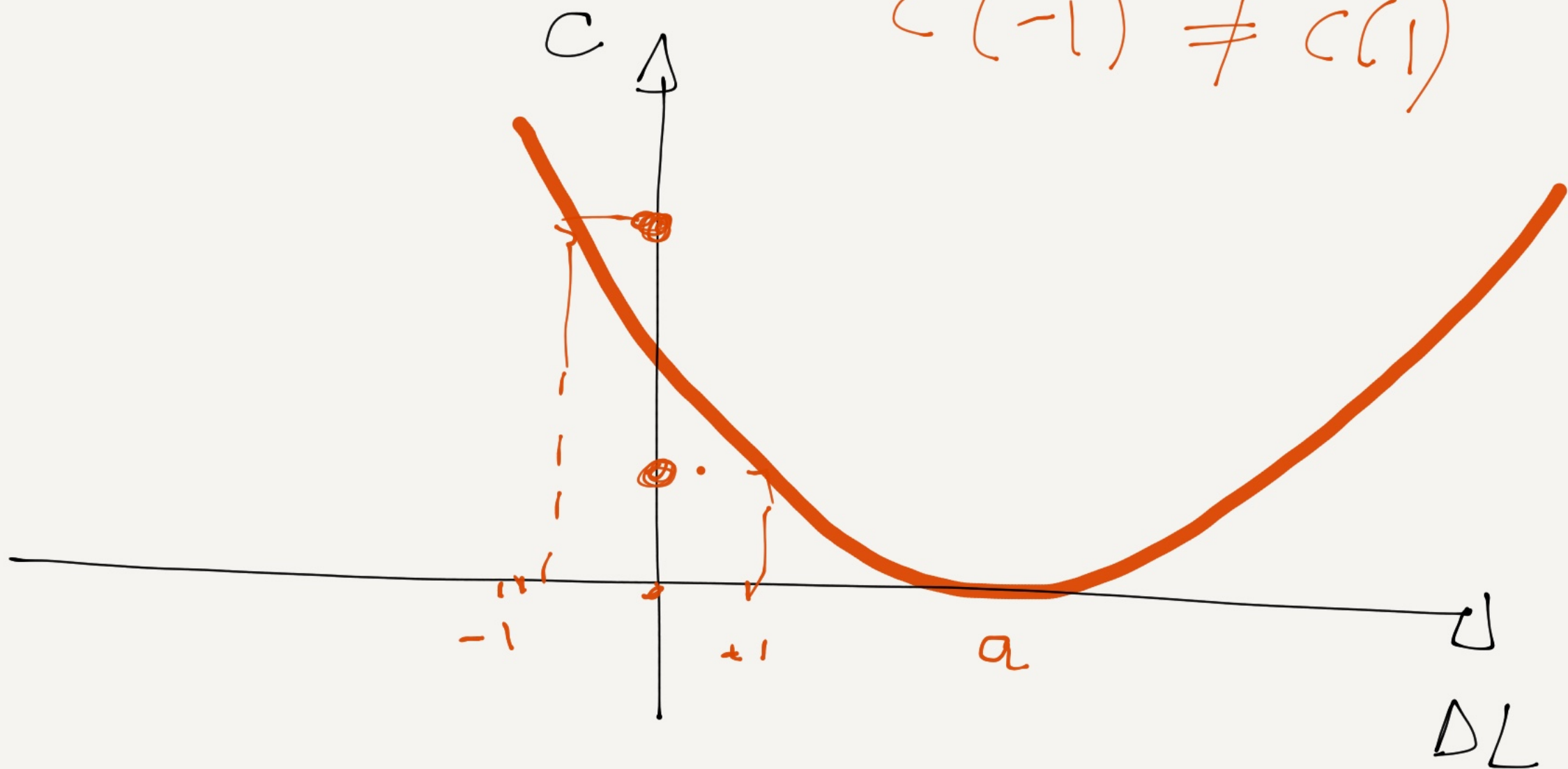
$$\tilde{y}_t = \rho \tilde{y}_{t-1} + \varepsilon_t^y$$



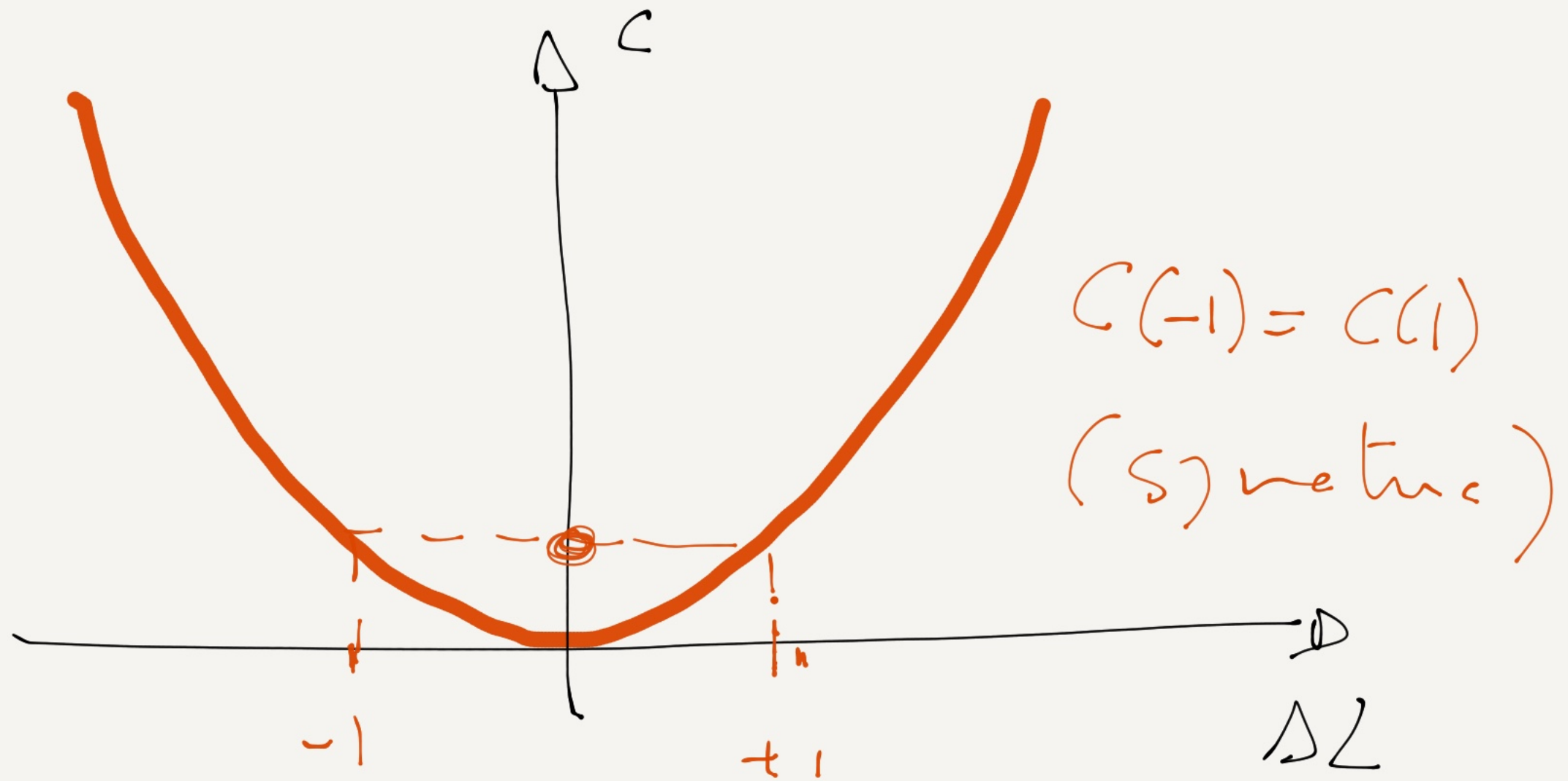
Adjusted cost to labor

$$C(\Delta L) = \frac{b}{2} (\Delta L - a)^2$$

$$C(-1) \neq C(1)$$



$$C(\Delta L) = \frac{b}{2} (\Delta L)^2 \quad (a=0)$$



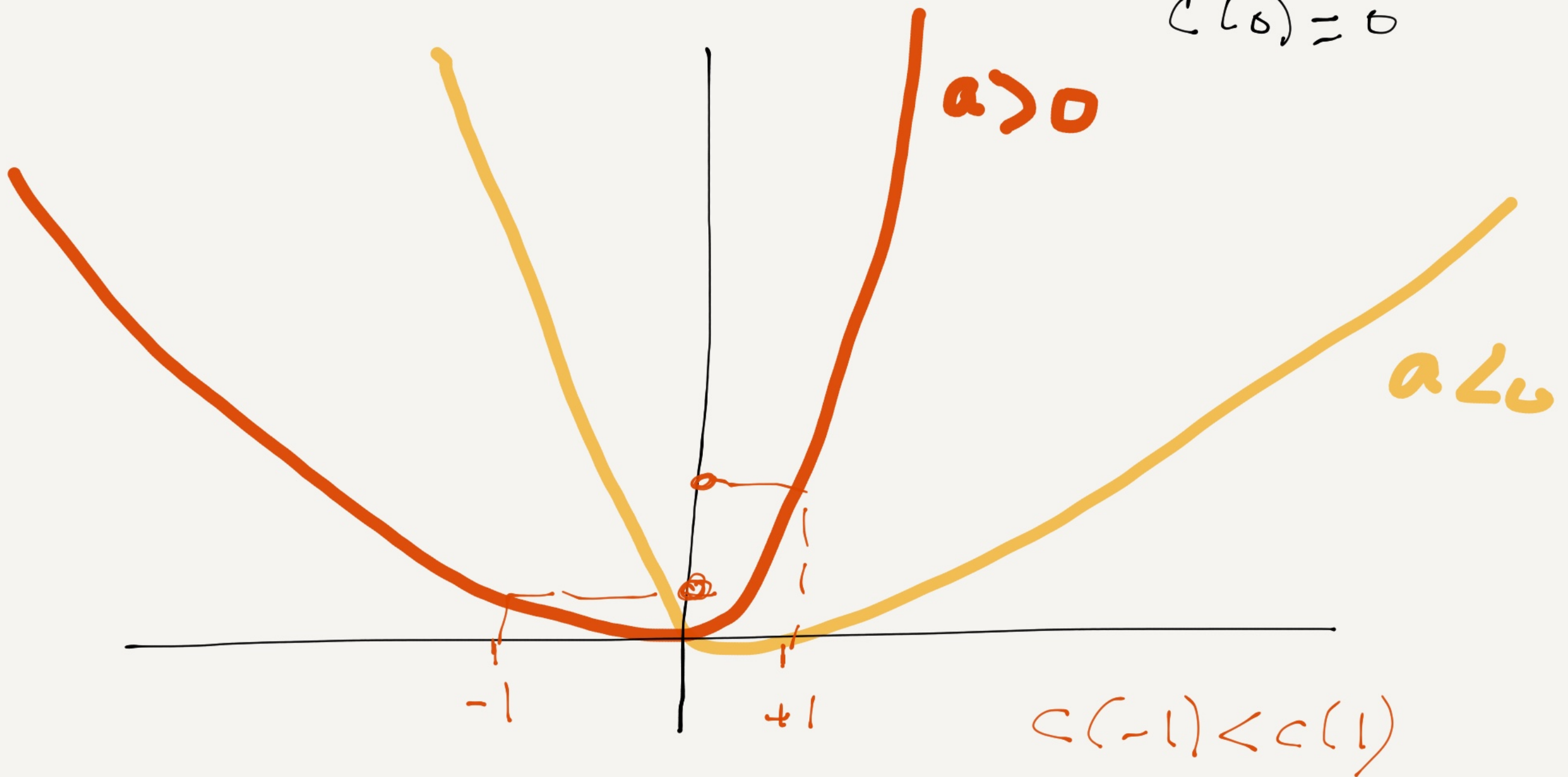
$$C(\Delta L) = -1 + e^{a\Delta L} - a\Delta L + \frac{b}{2} \Delta L^2$$

$$C(0) = 0$$

$$a > 0$$

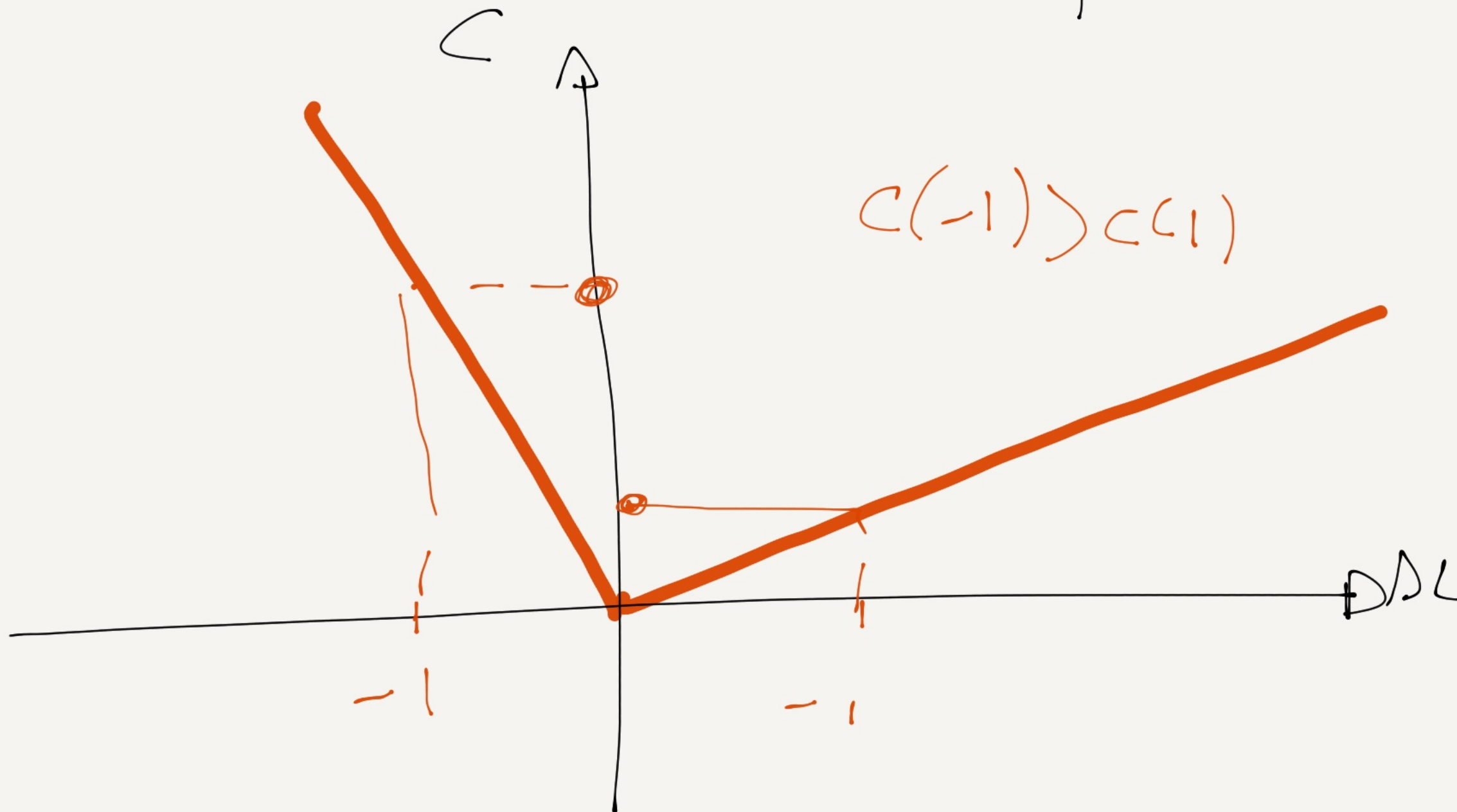
$$a < 0$$

$$C(-1) < C(1)$$



Linear costs

$$C(\Delta L) = \begin{cases} b_H \Delta L & \text{if } \Delta L > 0 \\ b_F \Delta L & \text{if } \Delta L < 0 \end{cases}$$



Linear + Fixed Costs

$$C(\Delta L) = \begin{cases} H + b_H \Delta L - y \Delta L > 0 \\ F + b_F \Delta L - y \Delta L < 0 \end{cases}$$

